

The Curiosity Museum

Product & Business Requirements Document

- A collection of 5–15 minute interactive explainers — a "digital museum" of niche, one-way topics*
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1. Executive Summary

You're describing a real, proven content category — it just doesn't have one agreed-upon name yet. People call it "explorable explanations," "interactive essays," or "visual scrollytelling." Individually built examples of this already exist and work (some as solo-creator side projects, some as venture-backed platforms), but **no one has packaged it as a browsing "museum" of unrelated niche topics** the way you're describing. That gap is your opening.

The strongest version of this product is **not** "everything in 3D." It's a curated collection where **each exhibit uses whatever medium best serves that specific topic** — sometimes a 2D scroll-driven diagram, sometimes a rotate-and-explore 3D object, sometimes a simple click-to-reveal sequence — unified by one consistent visual language, pacing, and quality bar. 3D is a tool you deploy selectively, not a blanket requirement, because full 3D for every topic will slow you down 3–5x for topics that don't actually need it.

2. Vision & Product Thesis

Vision: A single destination where someone with 10 spare minutes can walk into one "wing" of a museum and walk out actually understanding something they were curious about but never had explained well — human origins, how a click becomes a screen response, how planets compare, how a mechanical watch ticks, whatever.

Core thesis: People don't lack access to information; they lack a *format* that makes niche, slightly-technical curiosity feel like play instead of homework. Video is passive. Wikipedia is dense. Courses are too long. The gap is a format that is: interactive (you drive the pace), bounded (5–20 minutes, not an open-ended course), and visually delightful enough to share unprompted.

What makes it a "museum" and not a "course platform": no linear curriculum, no prerequisites, no grading. You wander in, explore one exhibit, and leave. That framing is also your competitive moat — see Section 4.

3. What This Is (and Isn't)

This is:

- A growing collection of standalone, self-contained interactive "exhibits," each 5–20 minutes
- Built for curious adults and teens, not students cramming for exams
- Designed to be screenshot-able, shareable, and individually linkable (each exhibit is its own URL/page)
- A mix of 2D scroll-driven pieces and select 3D/spatial pieces, chosen per-topic

This is not:

- A replacement for structured education (you already said this — keep repeating it in your own copy/marketing, it sets expectations correctly)
- A course platform with quizzes, certificates, or grading (that's Brilliant's and Articulate's territory — see below)
- A video library (video is a fallback medium here, not the primary one — interactivity is the whole differentiator)

4. Market Context: Who's Already Doing Pieces of This

You're not inventing the format — you're proposing a new *container* for a format that already has strong, independent proof points. Knowing them precisely tells you what to copy and what to avoid.

Project / Company	What it proves	What to borrow	What to avoid copying
Bartosz Ciechanowski (ciechanow.ski)	A solo creator can build museum-quality interactive explainers (bicycles, engines, GPS, gears, the Moon) and sustain it on ~\$3/month Patreon memberships	The "interactive essay" craft level, the slider/drag-to-explore interaction pattern, topic selection (everyday things nobody questions)	His pieces take months each — not sustainable for a wide, growing collection alone

Project / Company	What it proves	What to borrow	What to avoid copying
Nicky Case (ncase.me / Explorable Explanations)	"Explorables" work for abstract/social concepts too (game theory, voting systems, anxiety), not just physics	Design patterns for guided-vs-open exploration; treating interactivity itself as the teaching tool	Case's tools intentionally avoid monetization — you'll need a different revenue posture
The Pudding (pudding.cool)	Scroll-driven ("scrollytelling") data essays can go viral and build a large audience without a subscription paywall	Their scroll-trigger animation technique and "one clear visual argument per piece" discipline	They're ad/grant/institution-funded — not a direct model for your solo revenue plan
Brilliant.org	A freemium interactive-learning platform (STEM-only, problem-based) can scale to 10M+ users on a "first chapter free, subscribe for the rest" model, priced around \$13–25/month	The "unlock the full library" subscription structure, the daily-habit hook	Brilliant is a structured <i>course</i> platform with progression — the opposite of your open "wander in" museum framing. Don't structure your site like a curriculum
NASA Eyes on the Solar System	A single, extremely well-produced 3D spatial exhibit (rotate/compare/scrub time across planets and missions) can anchor an entire category on its own	Proof that one flagship 3D "wing" can be worth the extra build cost if the topic is inherently spatial	It's built by a full studio team — treat it as your aspirational <i>ceiling</i> for one flagship piece, not your baseline for every piece

Project / Company	What it proves	What to borrow	What to avoid copying
Corporate training tools (Articulate Storyline, Elucidat)	Branching, click-triggered, scenario-based interactivity is a mature, well-understood interaction grammar	The interaction <i>vocabulary</i> (branching choices, click-to-reveal layers, drag categorization) — reuse this grammar, aimed at curiosity instead of compliance	Their aesthetic (corporate, template-y, gamified with badges/points) is exactly what you want to avoid — you're going for "museum," not "onboarding module"

The actual gap: nobody has combined Ciechanowski's craft + Pudding's shareability + Brilliant's subscription mechanics + a "museum wings" structure that spans *any* topic, not just STEM or physics. That combination is your product.

5. Target Audience & Use Cases

- **The curious generalist** — reads Wikipedia rabbit holes for fun, watches Kurzgesagt/Veritasium, has 10 minutes and wants something better than another video
- **The parent/teacher looking for a "wow" moment** — wants a 10-minute classroom hook, not a full lesson plan
- **The shareable-content browser** — finds it via a tweet/reel of "wait, click on this, it's oddly satisfying," bounces in for one exhibit, may never come back — that's fine, it's still growth
- **The habitual return visitor** — the smaller group who bookmarks the site and checks back for new "wings"

Design decisions should optimize for the first two (acquisition/shareability) first, and the fourth (retention) second, once you have enough exhibits to justify a habit.

6. Core Product Concept: The Collection Model

Structure the whole site as a **museum floor plan**, not a folder of articles:

- **Wings** = topic domains (e.g., "The Body," "How Computers Think," "Deep Time," "Scale & Space," "Everyday Physics")
- **Exhibits** = individual 5–20 minute interactive pieces inside a wing
- **The Lobby** = homepage — a visual index of wings, designed to invite wandering (think a museum map, not a blog index)
- Each exhibit is a **standalone URL** — critical for shareability and SEO; someone should be able to land directly on "What Happens When You Click a Button" from a shared link or Google search without ever seeing the homepage

7. Interaction Design Patterns (Your Reusable "Verbs")

Rather than designing each exhibit from scratch, build a small library of reusable interaction patterns and mix them per topic. This is what keeps production sustainable as the collection grows:

1. **Scroll-scrub** — scrolling moves a timeline/animation forward and back (the button-click flow, evolution timelines, historical timelines)
2. **Rotate & inspect** — drag to orbit a 3D object, click hotspots for detail (organs, mechanical objects, planets)
3. **Zoom scale-shift** — pinch/scroll to zoom out from micro to macro or vice versa (cell → organ → body; Earth → solar system → galaxy)
4. **Click-to-reveal layers** — click a labeled part to peel back a layer and see what's underneath (skin → muscle → bone; code → OS → hardware)
5. **Slider/parameter play** — drag a slider and watch a system respond in real time (change voltage, change speed, change year)
6. **Branch & choose** — click a choice, see a different path/outcome (borrowed straight from corporate branching-scenario tools, repointed at curiosity instead of compliance)

Every exhibit you plan should map to 1–3 of these verbs before you start building. If a topic doesn't map cleanly to any of them, it's not ready for this format yet.

8. Flagship Exhibit Concepts (Worked Examples)

8.1 "What Happens When You Click a Button" (*your original idea — make this the flagship/first build*)

- **Verbs:** scroll-scrub + click-to-reveal layers
- **Flow:** user clicks a big button on screen → scrolling now drives a vertical journey: JS event → browser engine → OS call → CPU instructions/logic gates → electrical pulses →

response bubbling back up → pixel changes on screen

- **Medium:** primarily 2D (SVG/Canvas diagrams with GSAP ScrollTrigger), with a couple of simple 3D accents (e.g., a rotatable logic-gate diagram) — full 3D isn't needed for something this good as a 2D scrollytelling piece, so build it in 2D first
- **Why it's the right flagship:** novel angle, universally relatable ("everyone has clicked a button and never thought about this"), technically achievable in weeks not months

8.2 "The Body, Wing by Wing"

- **Verbs:** rotate & inspect + click-to-reveal layers
- **Flow:** a 3D rotatable human silhouette; click a system (circulatory, skeletal, nervous) to isolate it, then click individual organs to zoom in with detail cards
- **Medium:** 3D (React Three Fiber + a licensed or AI-generated anatomical model) — this is one of the few exhibits that genuinely justifies full 3D, since rotation and spatial relationship *is* the content
- **Note:** anatomical 3D assets are one of the few areas worth budgeting real money for; free/AI-generated models will look noticeably worse here than in other exhibits

8.3 "Scale: From Atom to Universe"

- **Verbs:** zoom scale-shift
- **Flow:** continuous zoom-out (or in) slider moving from a hydrogen atom to a human, to Earth, to the solar system, to the galaxy — same interaction spine as the classic "Powers of Ten" concept, updated with real comparative data
- **Medium:** 2D/2.5D is genuinely fine here (parallax layers), 3D optional for the planetary section only

8.4 "Deep Time: A Walk Through Earth's History"

- **Verbs:** scroll-scrub + slider/parameter play
- **Flow:** a horizontal or vertical timeline compressed to human scale, letting users "walk" from Earth's formation to now, with major events (life, extinction events, humans) appearing as they pass
- **Medium:** 2D scrollytelling, D3.js for the timeline logic

8.5 "Why Bicycles Don't Fall Over" / mechanical-systems series

- **Verbs:** rotate & inspect + slider/parameter play
- **Medium:** 3D, directly in the tradition of Ciechanowski's work — good candidates for AI-generated 3D models (Meshy/Tripo) since mechanical objects are exactly what those tools handle best today

8.6 "Origins: How We Got Here"

- **Verbs:** scroll-scrub + branch & choose (explore different hominin species)
- **Medium:** 2D illustrated scrollytelling with an optional 3D skull/skeleton viewer per species

Aim to launch with **one flagship (8.1) fully polished**, then add 2–3 more before opening any paid tier.

9. UX & Visual Design Philosophy

- **Minimal chrome, maximum content** — no persistent navbars, sidebars, or clutter inside an exhibit; let the interaction fill the screen, museum-exhibit style
 - **One idea per screen/scroll-stop** — never show two "verbs" competing for attention at once
 - **Consistent visual grammar across exhibits** — same typography, same color logic for "you are here" indicators, same transition style between exhibits — this is what makes it feel like *one museum* instead of a pile of unrelated demos
 - **Cinematic pacing over information density** — better to cut content than to cram; each exhibit should feel like it could end 30 seconds earlier than you want and users would still be satisfied
 - **Mobile-first interaction, not mobile-adapted** — a large share of shareable-content traffic (the exact growth engine this product depends on) will arrive on phones; every exhibit needs a real touch-first version, not a desktop demo with touch bolted on
 - **Always provide a 2D/lightweight fallback for any 3D exhibit** — for low-end devices and for accessibility
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10. Technical Architecture & Recommended Stack

Guiding principle: use the cheapest tool that does the job for each exhibit; reserve heavyweight 3D tooling for the exhibits that actually need it (Section 8 flags which ones).

Site shell / framework

- **Astro** (or Next.js if you want one unified framework) for the overall site — Astro's "islands" architecture is a strong fit here: most of each exhibit page is static/SEO-friendly, and only the interactive component hydrates as JavaScript. This matters a lot for a "collection of many niche pages" model, since each exhibit is effectively its own tiny SEO-optimized landing page.
- Tailwind CSS for styling

Animation / scroll-driven interaction (2D and 3D alike)

- **GSAP + ScrollTrigger** — the current standard for scroll-scrubbed timelines, works whether you're animating a plain SVG or a Three.js camera
- **Lenis** for smooth-scroll feel underneath GSAP
- **Framer Motion** for smaller UI micro-interactions (button states, card reveals)

3D layer (only for exhibits that need it)

- **Three.js via React Three Fiber (R3F)** for custom 3D scenes with full control (body, mechanical systems)
- `<model-viewer>` (Google's free, open-source web component) for simple single-object "rotate and inspect" cases that don't need a full custom scene — much less build time than R3F when all you need is "here's a 3D model, spin it around"
- **Spline** as a no-code shortcut for early prototypes or simpler hero scenes — it exports directly to a React component, so you can start there and later "graduate" a scene into full R3F if you need more control. Roughly \$9+/month once you outgrow the free tier.

3D asset creation & sourcing

- **Blender** (free) as your core 3D modeling tool
- **Blender MCP** ([ahujasid/blender-mcp](#) on GitHub, free/open-source) — connects Claude directly to a running Blender session via natural language, so Claude can build/adjust primitive-to-moderate scenes for you conversationally. Reliable for basic shapes, scene layout, and materials; still struggles with intricate organic geometry (like anatomy) — treat it as a fast-prototyping tool, not a replacement for a purchased/commissioned anatomical model.
- **AI text-to-3D generators** (Meshy, Tripo, Luma Genie) for quick asset generation — useful for mechanical objects and props; check licensing carefully, as most free tiers are either non-commercial or require attribution (CC BY), so budget for a paid tier once you're monetizing
- **Free real-world 3D scans**: Smithsonian Open Access and NASA's 3D Resources both publish free, usable glTF models of real artifacts and spacecraft — worth checking before generating or modeling something from scratch

Data-driven visuals

- **D3.js** for anything timeline- or scale-based (Deep Time, Scale exhibits)

Backend / accounts / payments *(build this later — you said defer monetization, and you're right to)*

- **Supabase** (free tier, Postgres + auth) for accounts and progress tracking when you're ready
- **Stripe** or **Lemon Squeezy** for payments — Lemon Squeezy is simpler for a solo creator selling subscriptions or one-time "unlock this wing" purchases

Hosting

- **Cloudflare Pages** (unlimited bandwidth on its free tier) is the safer default given 3D/media assets can be bandwidth-heavy; **Vercel** is the better developer experience if you build in Next.js, but its free "Hobby" tier technically restricts commercial use once you monetize, so budget for their Pro tier (\$20/mo) at that point

AI-assisted build workflow

- Use **Claude** (this chat, or Claude Code for larger builds) as your primary build partner — describe one exhibit at a time as a self-contained module/route so complexity doesn't compound across the whole site
- Use the **Blender MCP** connection specifically during 3D-asset sessions, separately from the main site-building sessions
- Keep each exhibit's code reasonably isolated (its own folder/component tree) — this lets you rebuild or replace one exhibit's approach later without risking the rest of the site

11. Content Production Reality Check

Be honest with yourself about production cost before committing to "everything in 3D":

- A well-built scroll-driven 3D web experience commonly takes in the rough range of **16–40+ hours** for a first build, even for an experienced developer — and that's for a *simple* one
- Ciechanowski-quality interactive essays take him **months** each, working solo
- Full-3D anatomical or spatial pieces (Section 8.2, 8.3) will cost meaningfully more than 2D scrollytelling pieces (8.1, 8.4) for the same "wow" payoff in many cases

Implication: plan a mix — most of your collection should be 2D/2.5D scrollytelling (cheap, fast, still highly interactive), with 3D reserved for the 2–3 exhibits per wing where spatial rotation is genuinely part of the content, not just decoration. This is also exactly why Ciechanowski and the Pudding, the two closest working proofs of this format, mostly *don't* use full 3D — cheap-to-produce 2D interaction, done well, reads as "high production value" to visitors.

12. Business Model & Monetization (*for when you're ready — kept separate from your MVP*)

Two viable postures, not mutually exclusive:

A. Solo-creator membership (near-term, lower risk)

- Free: most/all exhibits open, ungated — this is your growth engine (shareability only works if people can actually open the link you shared)
- Paid: a small monthly membership (\$3–8/month) for early access to new exhibits, behind-the-scenes/making-of content, or a few "deluxe" exhibits with extra depth — directly modeled on Ciechanowski's sustainable-but-modest Patreon approach
- This matches your own instinct to keep most things free and monetize "later" — it's the lower-commitment path

B. Freemium subscription platform (longer-term, higher ceiling)

- Free: first portion of each wing, or a rotating set of full free exhibits
- Premium: full collection unlocked, \$5–15/month — directly modeled on Brilliant's "first chapter free" mechanic, just applied to a museum-wandering format instead of a course sequence
- Adds real engineering cost (accounts, paywall, entitlement logic) — don't build this until you have enough exhibits that gating actually withholds something substantial

C. A later-stage option worth knowing about (not now): package sets of exhibits for schools or corporate L&D teams, the way Articulate Storyline content gets deployed into a company's learning platform — your "corporate training but better" comparison is actually a legitimate B2B angle once the free-audience side of the business is proven out.

13. Go-to-Market & Growth

- **Design every exhibit to be a self-contained, shareable unit** — a single clean URL, a good link-preview thumbnail, and a satisfying "aha" moment near the start, not the end. Single-topic visual pieces are what already spreads organically on X/Reddit/Hacker News in this exact content category.
 - **SEO the long tail** — niche "how does X actually work" queries have real search volume and comparatively little competition; each exhibit doubles as an SEO landing page if the framework (Astro) keeps pages fast and crawlable
 - **Target the specific communities that already love this format:** r/dataisbeautiful, r/explainlikeimfive, Hacker News, design-Twitter/X, r/Blender (for the 3D pieces specifically)
 - **A Product Hunt-style launch** works well for this category — visually strong, single-page interactive demos are exactly what performs there
 - **Educator outreach**, mirroring Brilliant's free-educator program, is a low-cost way to build credibility and word-of-mouth even before you have a B2B product
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14. Suggested Roadmap

MVP (weeks, not months):

- The Lobby (homepage) + 1 flagship exhibit, fully polished — recommend "What Happens When You Click a Button" (8.1)
- No accounts, no payments, no CMS — hand-build it

V1:

- 5–8 exhibits across 2–3 wings, including at least one 3D exhibit to prove out that pipeline
- Basic analytics to see which topics/exhibits actually get shared and revisited — let real behavior guide which wing you expand next, rather than guessing upfront

V2:

- Introduce the membership/paid tier (Section 12, Option A first — it's the lower-commitment path)
- Accounts + simple entitlement gating

V3 (only once the above is working):

- Consider the freemium-platform posture (Option B) or the B2B/education angle (Option C) — but only once free-tier traffic and engagement prove the core concept

15. Risks & Open Questions

- **Solo production bottleneck** is the single biggest risk — a wide "museum" needs many exhibits, and each one is a real production project. The reusable-interaction-pattern approach (Section 7) is your main defense against this.
- **3D performance on mobile** — test every 3D exhibit on a mid-range phone, not just your dev machine; have a 2D fallback ready
- **Differentiation** — strong, well-loved solo projects already exist in this exact space (Ciechanowski, Nicky Case). Your differentiation is breadth-as-a-collection and consistent "museum" framing, not out-producing any one of their individual pieces — don't try to beat Ciechanowski at his own single-piece depth early on
- **Licensing on AI-generated 3D assets** — confirm commercial-use terms before shipping any asset generated on a free tier
- **Monetization timing** — you've already made the right call to defer this; resist the urge to add payment complexity before the free collection has proven it can attract and hold an audience

16. Inspiration & Further Reading

- ciechanow.ski — interactive essays (bicycles, engines, GPS, the Moon)
 - ncase.me — Nicky Case's Explorable Explanations
 - pudding.cool — scrollytelling data essays
 - brilliant.org — freemium interactive STEM learning
 - science.nasa.gov/eyes — NASA's Eyes on the Solar System
 - github.com/ahujasid/blender-mcp — Blender MCP for Claude
 - modelviewer.dev — Google's `<model-viewer>` web component
 - gsap.com/docs/v3/Plugins/ScrollTrigger — scroll-driven animation
 - spline.design — no-code 3D design and web export
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- This document is a starting strategy, not a locked spec — treat Sections 8 and 14 as the parts most worth revisiting once you've built your first exhibit and have real user reactions to react to.*